



OPEN Deep learning-based phase demodulation for distributed acoustic sensor

Yiming Tang^{1,2,3}, Kewei Liu^{1,2}, Chen Liu^{1,2,3}, Haiyong Wu¹, Rugang Wang², Mengmeng Chen^{1,3}✉ & Fei Xu³✉

With the rapid advancement of deep learning, its applications in fiber optic sensing are expanding significantly, particularly in recognizing diverse scenarios where substantial progress has been made. However, the crucial task of obtaining demodulation results from raw data remains relatively underexplored. Accurate demodulation is essential for a deeper understanding of the physical processes in fiber optic sensing systems, enhancing measurement accuracy, and optimizing system performance. In this paper, a phase demodulation algorithm for distributed acoustic sensor (DAS) data processing was developed using deep learning framework. The proposed approach aims to replace the computationally intensive processes of traditional demodulation algorithms, such as the Hilbert transform, in specific scenarios. The proposed method enables faster acquisition of detailed demodulated data and provides an intuitive, real-time graphical display of the results. Experimental results demonstrate that the proposed method significantly enhances processing speed while maintaining accuracy comparable to traditional demodulation algorithms. Specifically, for a dataset comprising 2000 rows and 4500 columns (approximately 9 million data points), the response time is reduced from 2.62 to 0.1 s when processing data at a pulse frequency of 20 kHz over a sensing distance of 1.8 km.

Keywords Phase demodulation, Distributed acoustic sensor (DAS), Deep learning, Signal processing

To address the growing demand for safeguarding individual lives and protecting societal property resources, various monitoring technologies are developing rapidly, among which DAS in fiber optics sensing technology occupies an increasingly important position. DAS has been successfully applied to earthquake detection and early warning¹, bridge strain detection², boundary protection³, road traffic detection⁴, infrastructure testing of energy pipelines and communication networks⁵, etc. DAS provides an effective technical means for safety inspection, which can reduce maintenance costs with real-time inspection. This non-intrusive monitoring method has a wider measurement range and is more convenient than traditional inspection methods.

DAS achieves acoustic wave sensing by measuring the variation in optical phase caused by a change in the axial strain of an optical fiber. When an acoustic wave is applied to the sensing optical fiber, it causes the fiber strain in the axial direction, thus changing the phase of the Rayleigh scattered signal in the fiber. By demodulating the optical phase variation, the magnitude of the phase can be used to provide information about the intensity of the sound or vibration event, also the frequency and location of the event can be observed⁶.

Phase demodulation algorithm of DAS systems is an important technique used to acquire and analyze signal phase information, which is widely used in many fields. In 1993, Taylor et al. proposed a phase-sensitive optical time-domain reflectometer (ϕ -OTDR)⁷, which detects perturbations by measuring the intensity change of the backward Rayleigh-scattered light from a high-coherence light source, and it is mainly used for qualitative measurements. However, its performance is limited by low measurement accuracy and spatial resolution. Additionally, the lack of effective algorithms results in suboptimal noise filtering, susceptibility to environmental disturbances, and reduced reliability. In 2011, a phase demodulation OTDR based on digital coherent detection was proposed⁸, which realizes the digital demodulation of the phase difference between any two points. It can achieve high spatial resolution and accuracy while minimizing the impact of environmental noise on measurement results through advanced filtering and signal processing algorithms. After that, researchers designed various phase demodulation schemes, the scheme based on a 3×3 coupler can accurately demodulate

¹College of Electronic Engineering, Nanjing Xiaozhuang University, Nanjing 211171, China. ²College of Information Engineering, Yancheng Institute of Technology, Yancheng 224051, China. ³College of Engineering and Applied Sciences, and Collaborative Innovation Center of Advanced Microstructures, Nanjing University, Nanjing 210023, China. ✉email: chenmm@njzc.edu.cn; feixu@nju.edu.cn

the phase but requires three photodetectors, as well as the acquisition and processing of three signal channels⁹. The phase generation carrier (PGC) scheme¹⁰ offers high sensitivity and better noise immunity but requires additional circuitry, complex algorithms, and precise control of the modulation parameters. Similarly, the phase demodulation scheme based on I/Q demodulation and zero-difference detection^{11,12} introduces an alternative methodology. All these schemes enable distributed optical phase demodulation to address various application scenarios and requirements. Additionally, new technical solutions have emerged to enhance key performance metrics of DAS systems, including spatial resolution, frequency response bandwidth, noise reduction, and extended sensing distance. For instance, the ϕ -OTDR system based on three-pulse frequency division multiplexing¹³, achieves a higher signal-to-noise ratio (SNR). Likewise, ϕ -OTDR systems with continuous fading suppression, utilizing optimal tracking of multiple detection frequencies can predict and mitigate phase distortions. Moreover, ϕ -OTDR system employing chirped continuous light as the detection light¹⁴, improve both frequency response range and detection accuracy. With the increasing complexity of demodulation algorithms and the demand for real-time processing in diverse application scenarios¹⁵, DAS systems face challenges in handling massive data efficiently. Despite this, advancements in accelerating phase demodulation remain limited. This paper addresses these challenges by focusing on improving phase demodulation efficiency and reducing processing delays.

Traditional DAS data processing methods typically rely on a sequential execution of phase demodulation algorithms. For example, during the demodulation of modulated signals, the process involves initial filtering, followed by phase demodulation, and necessitates precise coefficient calculation, convolution operations, multiplication, addition, Fourier transforms, and other intricate steps to extract meaningful signals. This step-by-step computation approach is not only complex and time-consuming but also challenging to implement in real-time, leading to inefficient phase demodulation and making it difficult to meet the demands of large-scale data processing.

With the rapid development of AI technologies, deep learning applications in signal processing is becoming increasingly widespread. Its powerful nonlinear fitting capabilities and adaptive learning characteristics effectively handle complex nonlinear relationships in phase demodulation, improving accuracy, and automatically extract key features from input phase signals. Compared to traditional demodulation algorithms, deep learning techniques exhibit significant advantages in handling large-scale data. Despite this, research specifically focused on model training for DAS phase demodulation remains relatively scarce. This study aims to construct and train an efficient phase demodulation model, reducing time consumption and achieving real-time performance.

Principles

Phase demodulation in DAS is an algorithm for recovering the original phase information from a wrapped phase signal. When the optical fiber is in the receipt of external interference, the phase of light waves within it changes accordingly. Through the phase demodulation, the location and the vibration characteristics of the phase change can be determined. Commonly used phase demodulation algorithms include the digital coherent demodulation^{9,12,13}, PGC technology¹¹ and 3×3 coupler demodulation algorithm^{16,17}.

In this paper, digital coherent demodulation scheme is used to obtain the phase. The hardware scheme is given in Fig. 1a, a laser with a wavelength of 1550 nm and a power of 20 mW was used as the light source. The light was split into two parts by means of a 95:5 coupler. 95% of the light was modulated by an Acoustic Optic Modulator (AOM), which produced a 30 ns light pulse with a spatial resolution of 3 m and a repetition frequency of 20 kHz. Each optical pulse is amplified by an erbium-doped fiber amplifier (EDFA), and the amplified pulse is transmitted to the sensing fiber via an optical circulator, from which backscattered Rayleigh light from the sensing fiber is collected. The backscattered light and 5% of the light from the 1550 nm laser are simultaneously injected into a 50:50 optocoupler, the output of which is received by a balanced photodetector (30 kV/A transimpedance; 200 MHz bandwidth). Finally, a data acquisition (DAQ) card (sampling rate 250 MSa/s) collects the electrical signals from the detector output and analyzes the data via a personal computer.

The sensing fiber used in the experiment for training is 1.8 kilometers in length, and a 10-meter-long optical fiber is wound around a Piezoelectric ceramic Transducer (PZT) at a depth of 1630 to 1640 meters to serve as the simulated vibration source. To simulate a vibration signal, a 1 kHz signal with an amplitude of 5 V is applied to the PZT. The fiber-optic sensing data were collected several times, each time at a sampling frequency of 250 MSa/s, and the output fiber-optic sensing data were received through a Data Acquisition (DAQ) card and a computer for about 1 minute each time.

The flowchart of the traditional data processing method is expressed in Fig. 1b. First, the data acquired by DAQ card undergoes preprocessing operations including noise reduction and gain modulation, the gain modulation here specifically refers to the optical coherent gain achieved through heterodyne coherent detection. Next, phase information is extracted from the acquired data through three steps: Hilbert transform or phase-locked loop to extract the instantaneous phase, generation of the analytic signal, and arctangent calculation. The traditional Phase-Locked Loop (PLL) circuit is not directly employed. Instead, equivalent phase tracking and stabilization functions are achieved through digital signal processing algorithms. Then, phase unwrapping is performed. After this, the Fast Fourier Transform (FFT) data of all positions along the optical fiber are obtained. Finally, the intensity, location, and frequency of the vibration signal are output and displayed.

Consider the measured signal $s(t)$, the Hilbert transform of $s(t)$ is expressed:

$$H(s(t)) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{s(\tau)}{t - \tau} d\tau \quad (1)$$

Next, the analytic signal $z(t)$ is derived using:

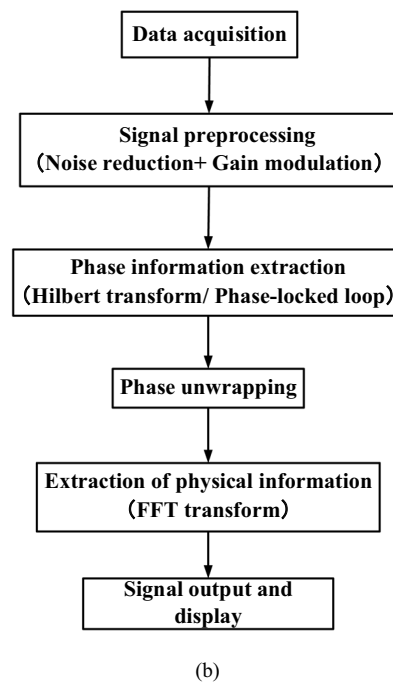
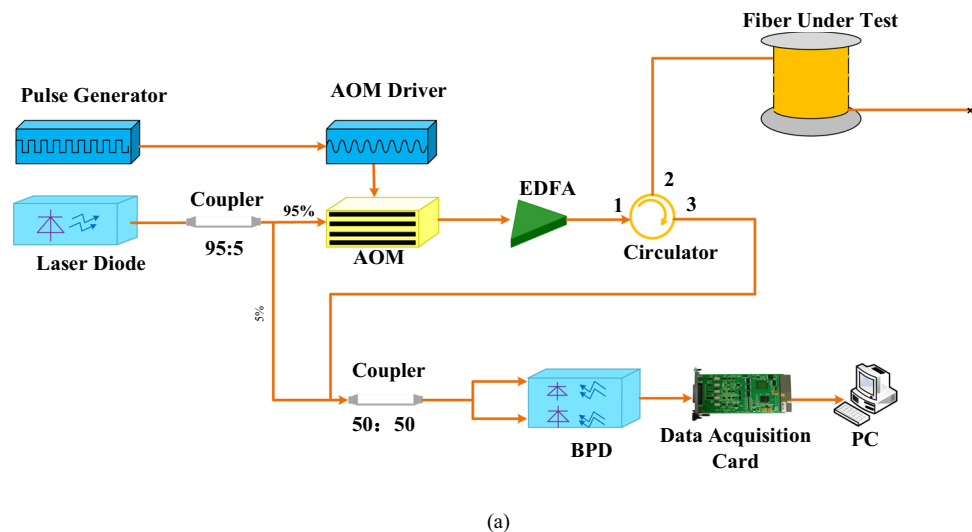


Fig. 1. The schematic diagram of the system's principle structure (a), and the flowchart of the traditional phase demodulation algorithm (b).

$$z(t) = s(t) + jH(s(t)) \quad (2)$$

Following this, the phase can be calculated as shown:

$$\varphi(t) = \arctan\left(\frac{H(s(t))}{s(t)}\right) \quad (3)$$

Subsequently, phase is unwrapped, allowing for the calculation of the phase variation in fiber using:

$$\Delta\phi(t) = \varphi_{unwrapped}(t) - \varphi_{unwrapped}(t - \Delta t) \quad (4)$$

where Δt represent the time interval of data acquisition. Finally, the useful information is extracted through FFT or difference calculation.

Training process

Preparation of datasets

The dataset consists of raw data collected by DAS during the experiment, along with the processed data obtained through traditional demodulation methods based on Hilbert transform.

The raw sensing data is one-dimensional time series got from the DAQ card.

During the data acquisition, the signal is divided into multiple segments at a rate of 20,000 segments per second, corresponding to the system's pulse repetition rate of 20 kHz. As the sampling rate is 250 MSa/s, each sample point represents approximately 0.4 meters of fiber length. The total sensing fiber length used for the training process is 1,800 meters, which corresponds to 4,500 sample points. Therefore, for each acquisition cycle, only the first 4,500 data points are retained, covering the entire sensing fiber. Every 0.1-second interval yields 2,000 segments, resulting in a dataset comprising $2,000 \times 4,500$ data points. The dataset was collected from different scenarios, including indoor, outdoor, no interference, and human interference conditions. From these scenarios, 100 data segments were selected as inputs for model training, with each data segment having a duration of 0.1 seconds. The data were collected under various combination conditions such as indoor, outdoor, no interference, and human interference, and 100 0.1 second data segments at different times were selected as inputs for model training.

Initially, the $2,000 \times 4,500$ points raw data of a single group is preprocessed and converted to normalized amplitude values through mathematical calculations. Next, the data is reshaped into a two-dimensional array with 2000 rows and 4500 columns to represent both time and space¹⁸. Then, it is necessary to remove the interference of the DC component:

$$y_{ij} = x_{ij} - \frac{\sum_{i=1}^n x_{ij}}{n} \quad (5)$$

y_{ij} is the filtered data in row i and column j , and x_{ij} is the original data, and n is the total number of rows in the column.

Subsequently, digital averaging is applied to every five columns of the filtered 2000-row by 4500-column data matrix to reduce random noise interference. Each column is then processed using the Hilbert transform algorithm, resulting in a corresponding target data matrix of 2000 rows by 900 columns. In other words, every five columns of the original 2D data correspond to one column in the target matrix. For the purpose of model training, the original variables and target variables are combined into a single dataset with a shape of 1800,000 rows by 6 columns, where the first five columns represent the original variables and the last column represents the target variable. The data conversion process is expressed in Fig. 2:

After integrating a large amount of data according to the specified dataset format, 70% of the data is used as the training set, 20% as the validation set for model tuning, and the remaining 10% as the test set to evaluate the generalization ability of the trained model. All data are collected in chronological order, and there are no overlapping time segments among the training/validation/test sets, which ensures their independence.

Finally, normalize all feature data using the statistical metrics (such as mean and variance) of the feature data in the training set:

$$X_{norm} = \frac{X - \mu}{\sigma} \quad (6)$$

Model training

A neural network model is an artificial intelligence model inspired by biological nervous systems. It consists of a large number of interconnected simple units (neurons), which interact according to specific structures and rules to achieve complex information processing and learning functions^{19,20}. For time-series signals, to mitigate overfitting, improve the approximation of the multi-class nonlinear relationship, and enhance the fault tolerance of the model, the BP neural network algorithm was selected. The BP neural network is known for its strong nonlinear mapping and generalization abilities, and its capacity to learn and store a large number of input-output mapping relationships.

The BP neural network, also known as a fully connected neural network, is composed of three types of layer structures: an input layer, hidden layers, and an output layer. Data is fed into the neural network through the input layer, processed through a series of intermediate (hidden) layers, and finally generates outputs. The training process of a neural network is typically achieved through the backpropagation algorithm, which adjusts the connection weights between neurons to enable the network to adapt to and learn the patterns of input data^{21–23}. Initially, the BP neural network was designed with 5 input neurons, one hidden layer (determined through testing), 15 neurons in the hidden layer, the ReLU activation function, and the mean square error (MSE) as the objective function for optimization. The Adam optimization algorithm was employed to adjust the network parameters and minimize the loss function.

There are three main scenarios for using forward and backward propagation: batch gradient descent, stochastic gradient descent, and mini-batch gradient descent. Batch gradient descent processes all samples at once, which is computationally expensive and slow. Stochastic gradient descent processes one sample at a time and updates parameters with each sample, providing faster results but with poorer convergence, potentially causing oscillations around the optimal point. Mini-batch gradient descent, which divides the samples into batches, balances the two approaches, updating parameters in batches.

Shuffle the divided training set before training, and mini-batch training was employed, where data were loaded in random order for each batch. Each batch consisted of 128 data points, and the network parameters were updated after every batch to improve training efficiency and reduce overfitting. Additionally, the learning

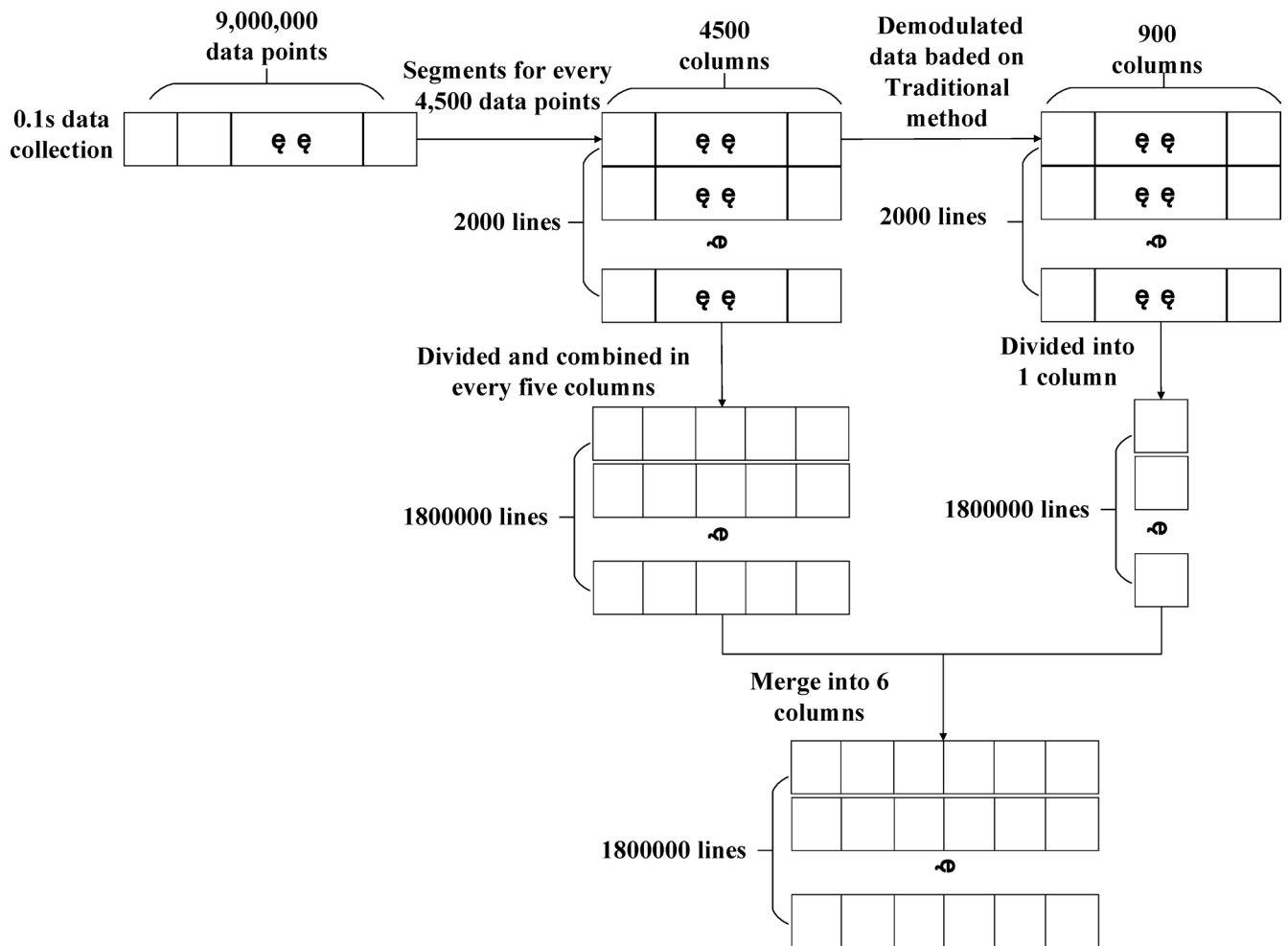


Fig. 2. The data conversion process.

rate was dynamically adjusted during training to prevent gradient explosion or vanishing gradients. A learning curve was plotted to observe accuracy trends in both the training and validation sets over training rounds, helping to detect overfitting or underfitting.

The final model saved after training was not necessarily the best, so models were compared based on the loss function values from the training set, and the model with the lowest loss was saved. However, this model may not perform well on other datasets. Therefore, after each model was trained, a separate validation set was introduced without backpropagation to evaluate the model, and the model with the lowest test set loss was saved.

Evaluation and validation of the model

This model runs on a CPU. The PC is equipped with an Intel Core i7-12700K processor, 32GB of RAM, a 300GB hard disk, and the Windows 10 operating system, with the QT1144 DAQ card configured. The objective of this work is to enhance the data processing speed while maintaining accuracy. Therefore, the primary evaluation metrics are error and running time.

The hidden neurons are used as an example to compare the difference in accuracy and runtime of the model for different numbers of cases. The running time here is the time taken by the model to analyze a single data set of 2000 rows and 4500 columns. Training is performed using the Mean Squared Error (MSE) of the validation set, and finally evaluated by the Mean Absolute Percentage Error (MAPE) of the test set. The results are expressed in Table 1.

For the error function used in training this model, unlike the binary cross-entropy (BCE) loss function typically applied for classification tasks, this study employs the MSE function, which is widely used in regression tasks, for model training. MSE calculates the average of the squared differences between the predicted and true values. One key advantage of MSE is that it assigns greater weight to larger errors, making it more sensitive to outliers. The formula is as follows:

$$MSE = \frac{1}{N} \sum_{i=1}^n (x_i - y_i)^2 \quad (7)$$

Sequences	Treatment	Hidden neuron	MSE	MAPE (%)	Running time(s)
1	BP neural network	5	0.15	1.38	0.06
2	BP neural network	10	0.02	0.50	0.07
3	BP neural network	15	0.004	0.22	0.08
4	BP neural network	20	0.002	0.19	0.10
5	BP neural network	25	0.002	0.17	0.14
6	Traditional demodulation				2.62

Table 1. Effect of different number of neurons on accuracy and running time.

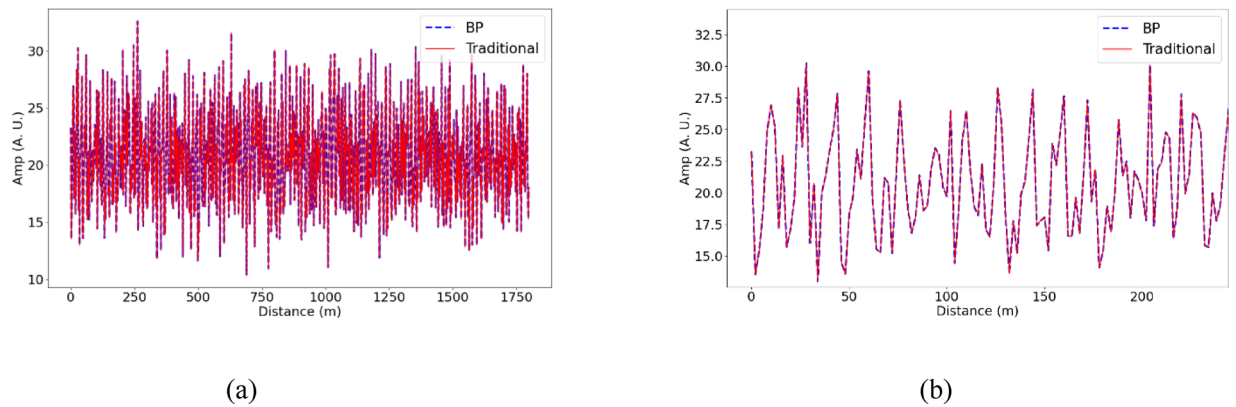


Fig. 3. Comparison diagram of results from two different data processing methods, (a) Data along the entire fiber, (b) partial enlargement of (a).

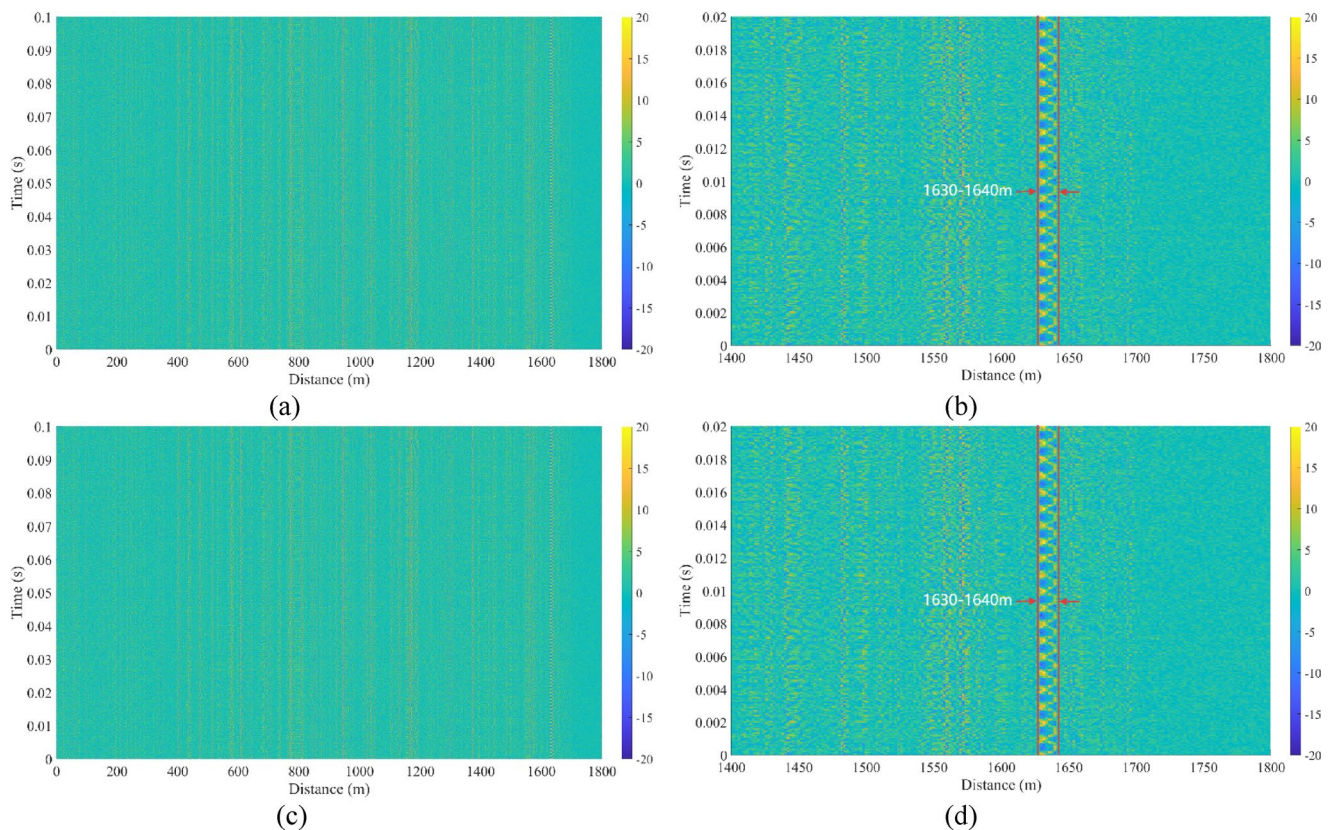


Fig. 4. 3D top view, (a) overall view of the data got by using traditional data processing method, (b) partial enlargement of (a), (c) overall view of the data got by trained BP neural network model, (d) partial enlargement of (c).

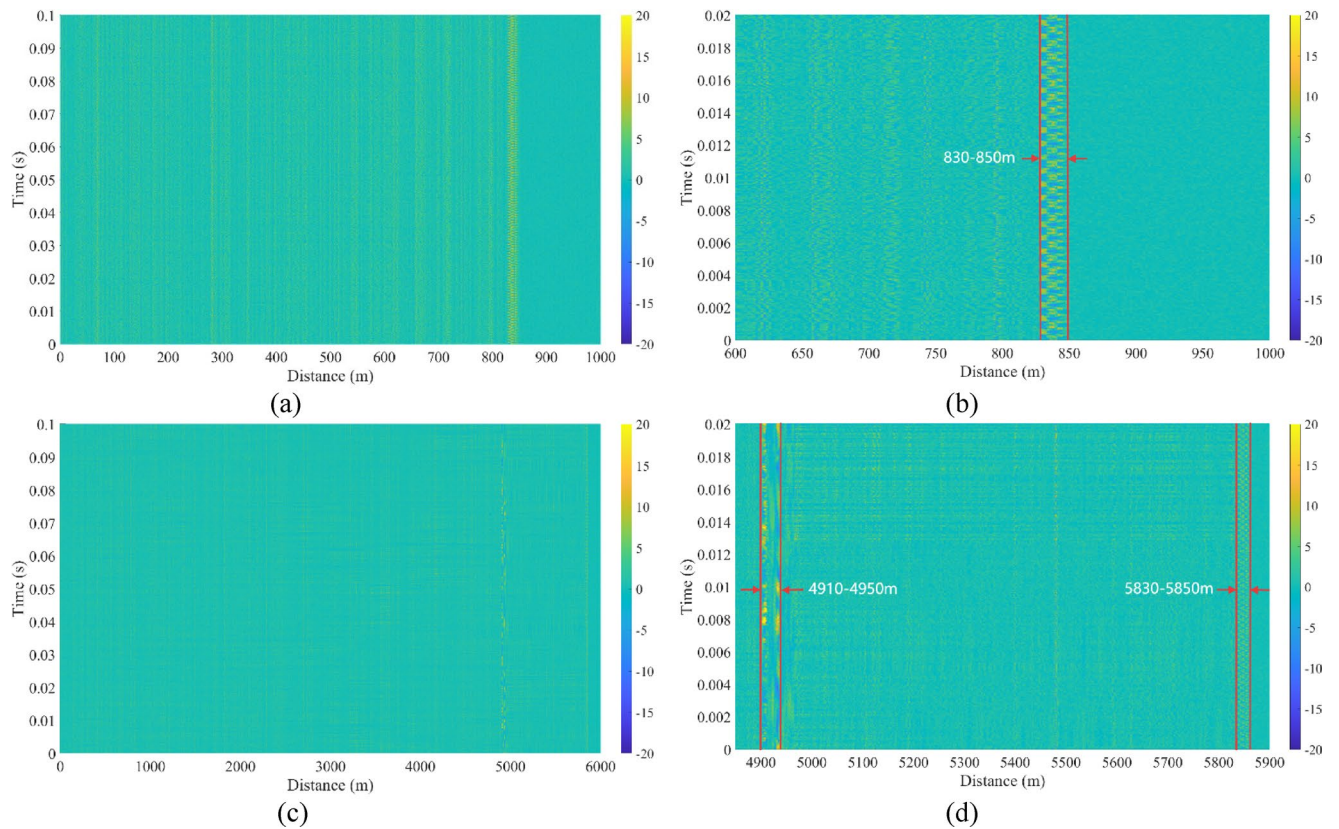


Fig. 5. Disturbance diagram along the sensing fiber, (a) sensing fiber with a length of 1 km, (b) partial enlargement of (a), (c) sensing fiber with a length of 6 km, (d) partial enlargement of (c).

where N is the number of training samples, and x_i is the predicted value of the i_{th} output, and y_i is the true value corresponding to the i_{th} output.

The error rate of the BP neural network in processing this fiber optic sensing data is very low, with the MSE being approximately 0.004 when compared to the traditional method.

Whereas the MAPE was used to calculate the error in the actual test, the

$$MAPE = \frac{1}{N} \sum_{i=1}^n \left| \frac{x_i - y_i}{y_i} \right| * 100\% \quad (8)$$

where N , x_i , and y_i represent the same variables as in Eq. (7). MAPE expresses the error in the form of percentage, which can intuitively reflect the degree of deviation of the predicted value relative to the true value and is comparable for data of different magnitudes. Experimentally, the percentage error can be obtained in the range of 0.2–0.4%.

After the training is completed, a line graph is used to visually determine the error of the data processed by the traditional scheme and the BP neural network. The raw data of a section of 1800m long optical fiber at a certain point in time is taken and processed by two methods respectively. The data processed by the traditional phase demodulation scheme (indicated by the red curve) and the data processed by the BP neural network (indicated by the blue line) are plotted in a graph for comparison. As shown in Fig. 3 below, the two curves almost completely overlap, indicating a high prediction accuracy.

Discussion

The experimental environment for this data collection was conducted indoors, where temperature and humidity conditions were controlled normally to ensure the stability of the experimental results.

Finally, the predictions are compared with the real data by converting them into a three-dimensional model, with time, space, and magnitude as the three axes, the top view of this 3D drawing and its partial enlargement are shown in Fig. 4 below.

Based on the results of Fig. 4, it can be found that the prediction map in Fig. 4d and the target in Fig. 4b are the same at 1630 m in the space noted by the red box, there is an obvious stripe-like mutation, and this anomalous location is the analogue perturbation of the sinusoidal voltage signal generated by the function generator.

Comprehensively analyzing the results of a lot of experiments, the 15 hidden neuron is the most cost-effective. At this point, the error is small and within an acceptable range. Increasing the number of hidden neurons further

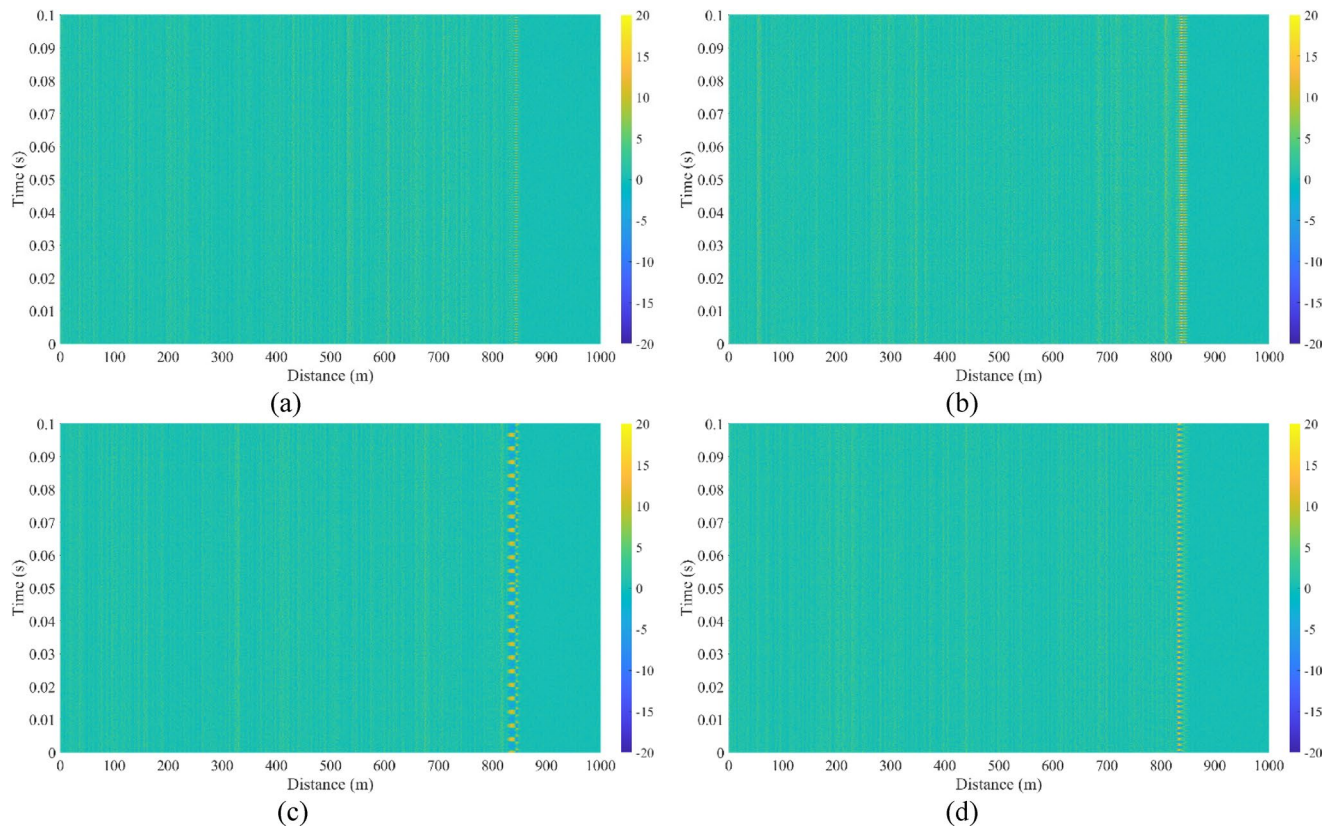


Fig. 6. Responses to vibrations with varying signal amplitudes and frequencies along the 1 km fiber, (a) Frequency of 1 kHz and amplitude of 1 V, (b) Frequency of 1 kHz and amplitude of 2 V, (c) Frequency of 200 Hz and amplitude of 5 V, (d) Frequency of 500 kHz and amplitude of 5 V.

does not significantly reduce the error, while the consumed time will rise rapidly. For the processing of collected raw data, in terms of accuracy, the difference between BP neural network and conventional methods demodulated data is only less than 1% (based on the data of Table 1), and the time required for BP neural network to process the 2D data of 2000*4500 once is only 0.1s on average, while conventional method require 2.62s. The BP neural network is greatly exceeded in terms of efficiency, which is crucial for the real-time processing of optical fiber signals.

The model's input data is a reshaped matrix signal¹⁸, containing data corresponding to all physical positions along the optical fiber. To validate the model's generalization capability, original data collected under varying fiber lengths, signal amplitudes, signal frequencies, and waveforms were processed by the model without retraining, while keeping other parameters constant. Experiments results show that the model of the BP neural network shows generalization potential in other unseen fiber path tests, the resulting outputs from the model are presented in Figs. 5, 6, and 7, respectively.

The data in Fig. 5a and b were obtained from experiments conducted on a 1 km-long optical fiber. A PZT was installed at the 830–850 m position along the fiber to simulate a signal source. In Fig. 5c and d, the data were collected from a 6 km-long fiber with two vibration sources applied: one located between 4910 and 4950 meters, and the other between 5830 and 5850 meters. The experimental results demonstrate that the disturbance signals at all the aforementioned locations were successfully detected. The MAPE of the two figures in Fig. 5 was 0.20% (for (a) and (b)) and 0.26% (for (c) and (d)), respectively.

Figures 6 and 7 presents the experimental results obtained from a 1 km-long optical fiber, with the vibration applied at a position between 830 m and 850 m. The figure examines different types of input signals. Figure 6a and b both use a signal frequency of 1 kHz, but with different amplitudes: 1 V and 2 V, respectively. In contrast, Fig. 6c and d use a fixed amplitude of 5 V but differ in frequency, being 200 Hz and 500 Hz, respectively. The results clearly demonstrate that the system responds significantly to all signal types. Although the response in Fig. 6a is the weakest, the vibration signal can still be effectively detected using a threshold-based identification method. The MAPE of the figures in Fig. 6 was 0.22%, 0.19%, 0.22% and 0.23%, respectively.

In Figs. 7a illustrates the system's response to a 1 kHz triangular wave, while Fig. 7b shows the response to a sweep signal ranging from 1 to 10 kHz. The MAPE of the figures in Fig. 7 was 0.20% and 0.24%, respectively. The results demonstrate that the system also exhibits strong responsiveness to such specialized input signals.

Through a series of diverse experiments, along with validation in other application scenarios conducted by the authors, the proposed model has demonstrated a level of generality and robustness in signal demodulation comparable to that of traditional phase demodulation algorithms. It exhibits high efficiency when applied to

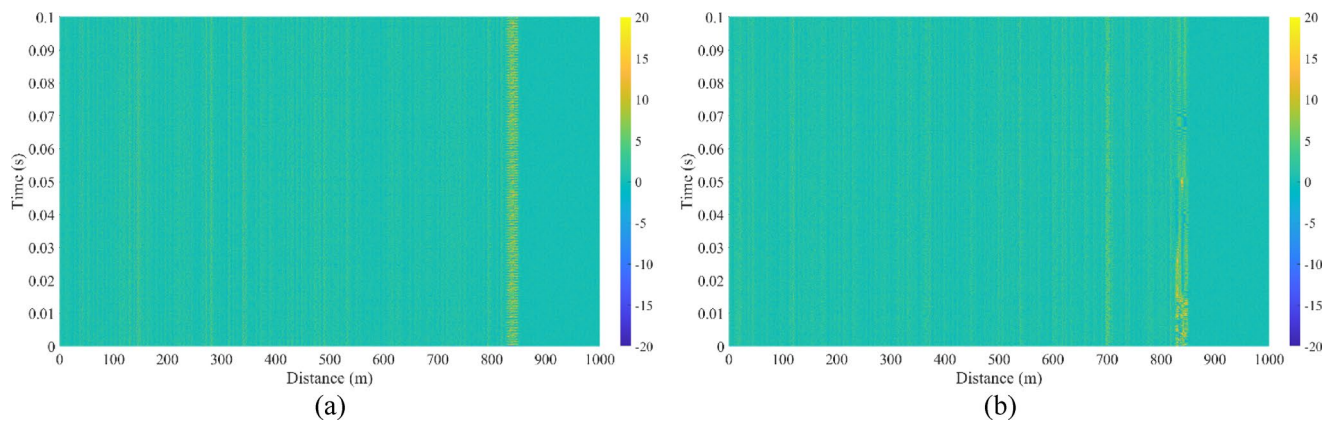


Fig. 7. Responses to vibrations with varying signal amplitudes and frequencies along the 1 km fiber, (a) triangular wave 5 V @ 1 kHz (b) swept sinusoidal wave from 1 to 10 kHz.

signal demodulation tasks in complex environments. The highly compact architecture of the BP neural network significantly reduces computational complexity and parameter count, thereby enhancing its performance in vibration signal recognition and data demodulation applications.

Conclusions

In this study, a BP neural network model is introduced, capable of automatically identifying and demodulating data, as well as determining the location of anomalous fibers through training on large volumes of fiber optic data. Compared to the traditional phase demodulation algorithm, the optimized BP neural network model significantly reduces computation time when processing fiber data of the same size. This allows for more timely access to demodulation results and anomaly location information, which is critical for real-time monitoring and rapid response systems. Additionally, for fiber optic data of varying types, sizes, and features, the BP neural network model, with appropriate training and tuning, maintains strong demodulation performance and anomaly detection capabilities. This highlights its impressive generalization ability across diverse fiber data sets.

Data availability

The dataset used and analyzed in the current research are available from the corresponding authors upon reasonable request.

Software declaration

The 3D top-view images presented in Figs. 4–7 were generated using MATLAB R2023a (© 1984–2023 The MathWorks, Inc.), licensed under the Nanjing University Campus-Wide Agreement. Access note: Installation resources are available at institutional repositories (the URL link: https://download.nju.edu.cn/matlab/R2023a_U1/matlab_R2023a_U1_Windows.iso).

Received: 23 December 2024; Accepted: 28 July 2025

Published online: 13 August 2025

References

- Marra, G. et al. Ultrastable laser interferometry for earthquake detection with terrestrial and submarine cables. *Science* **361**, 486–490 (2018).
- Rao, Y., Wang, Z., Wu, H., Ran, Z. & Han, B. Recent advances in phase-sensitive optical time domain reflectometry (Φ -OTDR). *Photonic Sens.* **11**, 1–30 (2021).
- Wu, H. et al. in *23rd International Conference on Optical Fibre Sensors*. 1285–1288 (SPIE).
- Wang, Z. et al. in *2017 25th Optical Fiber Sensors Conference (OFS)*. 1–4 (IEEE).
- Duckworth, G. L. & Ku, E. M. in *Sensors, and command, control, communications, and intelligence (c3i) Technologies for homeland security and homeland defense XII*. 90–103 (SPIE).
- Zhou, F. et al. A real-time phase processing system for phase sensitive optical time domain reflectometer. *Rev. Sci. Instrum.* **94**, 014710 (2023).
- Taylor, H. F. & Lee, C. E. Apparatus and method for fiber optic intrusion sensing (1993).
- Pan, Z. et al. in *2011 Asia Communications and Photonics Conference and Exhibition (ACP)*. 1–6 (IEEE).
- Masoudi, A., Belal, M. & Newson, T. A distributed optical fibre dynamic strain sensor based on phase-OTDR. *Meas. Sci. Technol.* **24**, 085204 (2013).
- Fang, G., Xu, T., Feng, S. & Li, F. Phase-sensitive optical time domain reflectometer based on phase-generated carrier algorithm. *J. Lightwave Technol.* **33**, 2811–2816 (2015).
- Wang, Z. et al. Coherent Φ -OTDR based on I/Q demodulation and homodyne detection. *Opt. Express* **24**, 853–858 (2016).
- Shan, Y. et al. Performance optimization for phase-sensitive OTDR sensing system based on multi-spatial resolution analysis. *Sensors* **19**, 83 (2018).
- Zabihi, M. et al. Continuous fading suppression method for Φ -OTDR systems using optimum tracking over multiple probe frequencies. *J. Lightwave Technol.* **37**, 3602–3610 (2019).
- Jiang, J. et al. Continuous chirped-wave phase-sensitive optical time domain reflectometry. *Opt. Lett.* **46**, 685–688 (2021).

15. Wang, Y. et al. Long-range chaotic Brillouin optical correlation domain analysis with more than one million resolving points. *Adv. Photonics Nexus* **2**, 036011–036011 (2023).
16. Chen, M., Masoudi, A. & Brambilla, G. Performance analysis of distributed optical fiber acoustic sensors based on ϕ -OTDR. *Opt. Express* **27**, 9684–9695 (2019).
17. Chen, M., Masoudi, A., Parmigiani, F. & Brambilla, G. Distributed acoustic sensor based on a two-mode fiber. *Opt. Express* **26**, 25399–25407 (2018).
18. Chen, M. et al. Distributed vibration sensor with a high strain dynamic range by harmonics analysis. *Opt. Laser Technol.* **159**, 109006 (2023).
19. Zhang, L., Wang, F., Sun, T. & Xu, B. A constrained optimization method based on BP neural network. *Neural Comput. Appl.* **29**, 413–421 (2018).
20. Li, J., Cheng, J.-h., Shi, J.-y. & Huang, F. in *Advances in Computer Science and Information Engineering*, Vol. 2. 553–558 (Springer).
21. Jia, H. et al. A k-nearest neighbor algorithm-based near category support vector machine method for event identification of ϕ -OTDR. *IEEE Sens. J.* **19**, 3683–3689 (2019).
22. Wang, X., Liu, Y., Liang, S., Zhang, W. & Lou, S. Event identification based on random forest classifier for Φ -OTDR fiber-optic distributed disturbance sensor. *Infrared Phys. Technol.* **97**, 319–325 (2019).
23. Kandamali, D. F. et al. Machine learning methods for identification and classification of events in ϕ -OTDR systems: a review. *Appl. Opt.* **61**, 2975–2997 (2022).

Author contributions

Conceptualization, M.C. and H.W.; methodology, M. C. and Y. T.; software, Y. T.; validation, M.C. and Y. T.; formal analysis, M.C.; investigation, Y. T.; resources, H.W., R. W. and F. X.; data curation, Y. T.; writing—original draft preparation, Y. T.; writing—review and editing, M.C., H.W., R. W. and F. X.; supervision, F. X.; funding acquisition, M. C. and F. X. All authors have read and agreed to the published version of the manuscript.

Funding

The project is supported by the National Natural Science Foundation of China (62105157), Jiangsu Province post-doctoral Science foundation (2021K228B), and Natural Science Foundation of Jiangsu Province (BK20211014).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to M.C. or F.X.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025